

RICE HUSK ASH (RHA) BASED CONCRETE: WORKABILITY AND COMPRESSIVE STRENGTH WITH DIFFERENT DOSAGES AND CURING AGES

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Received: 15.01.2022 / Accepted: 11.02.2022 / Revised: 07.03.2022 / Available online: 31.05.2022

DOI: [10.2478/jaes-2022-0016](https://doi.org/10.2478/jaes-2022-0016)

KEYWORDS: rice husk ash, concrete, cement, workability, compressive strength.

ABSTRACT:

To reduce the consumption of cement in construction industry has been a hot area of consideration now days due to high evolution of CO₂ gases during its production. Since decades efforts are made to replace cement with cementitious materials; minerals, agricultural, industrial by products etc. A few are fly ash, ground granulated blast furnace slag, silica fume, rice husk ash etc. These cementitious materials are often adopted as partial replacement of cement. However, their effectiveness and suitability as cementitious material depends upon their properties and the source of the production. Pakistan is an agricultural country and one of the major crops is Rice crop. It produces a substantial amount of rice husk during the processing of the rice. A substantial amount of Rice Husk Ash (RHA) is produced on the burning of rice husk used as fuel in the rice mills while generating steam for parboiling process of rice grains. This ash causes the environmental problems also if not dumped properly. RHA is believed to have siliceous properties which may be used as supplementary cementitious material in concrete. As the suitability of supplementary cementitious materials is dependent upon the source of rice husk, temperature of burning of husk, its fineness and other properties. This study is focused on to determine the effect of rice husk ash produced by burning of the rice husk obtained from the local rice mills. Experimental investigation carried includes the determination of workability and compressive strength of concrete with different dosage of RHA from 5% to 30% with an increment of 5% tested at 1day, 3days, 7days and 28 days of the age of concrete. The results are compared with corresponding concrete without RHA and tested at the same ages. The results show the reduction in both the parameter; workability and compressive strength due to RHA particularly at its higher dosage beyond 10% by weight of cement.

1. INTRODUCTION

Because of its structural stability and strength, the concrete is a popular building material used in the structures. Concrete is a mixture of cement, aggregates, and water. It plays a vital role in the economic development of any country. It is a durable material and possesses an unquestionable quality of life. The use of concrete in the construction industry has been essential due to its unmatched performance compared to other materials. The manufacture of concrete in terms is continually growing manifold than the other materials being used for the construction. This is primarily due to human migration from rural to urban areas. However, this causes the massive consumption of the common ingredients normally obtained from natural resources. On the other hand, huge consumption of OPC results in greenhouse gas emissions during the cement manufacture, adversely affecting the environment. Altogether, the advancement in industrialization and agriculture has provoked another issue that also affects the environment. The problem is the generation of waste as industrial by-products, agricultural wastes etc. The dumping of this waste needs a vast

area and if not appropriately dumped results in a hazardous effect on the environment, thereby badly affecting human health. For decades, efforts are being made by the construction industry and the researchers to find the ways and means to reduce the consumption of OPC in concrete and use these wastes as useful construction ingredients in concrete in general and as a binder in particular.

Hence the researchers suggest the use of green materials which contain pozzolanic properties. Several green resources, such as fly ash, rice husk ash, blast furnace slag, silica fume, and damaged building materials have been investigated for partial substitution of cement. Recent technical advancements have demonstrated that these materials are useful materials that can create various valuable products. The cement and concrete industry has increasingly used mineral admixtures since the mid-twentieth century. Partial cement replacement addresses the growing demand for cement and concrete. This research focuses on studying the possibility of the partial replacement of cement with Rice Husk Ash (RHA). Rice husks are common agricultural byproducts. These are the shells generated during

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the de-husking of paddy rice. It accounts for 20% of the world's paddy production of 500 million tonnes (Bhanumathidas and Mehta, 2004).

Pakistan is an agricultural country having a wide variety of crops. Amongst the major crops, one is the Rice. This crop causes the generation of an adequate husk during the processing. Rice is a major Kharif crop in Pakistan. Pakistan produced 7.4 million tonnes of rice in 2020, ranking 10th among rice-producing countries. The use of industrial by-products as partial replacement of cement is recognized for energy and cost-saving. Further, the use of by-product having cementing properties might reduce the usage of cement quantity. Several by-product materials can be used as a replacement for cement. Among those, Rice Husk Ash (RHA) is a potential material as it contains a high amount of Silica (Bui et. al., 2005; Mehta, 1977).

This motivated to undertake this experimental study to understand the possibilities of "Rice Husk Ash" as a binder in the manufacture of concrete. The study is experimental in its nature, focusing on using RHA as a binding agent in the concrete. Standard concrete of 1:2:4 ratios were used throughout the study with a fixed w/c ratio being 0.5. In all seven batches were cast including ordinary concrete. Remaining six batches include RHA content from 5% to 30% with increment of 5%. OPC was used as cement and was reduced by weight of binder equal to the RHA percentage in respective batch. Two tests were carried out i) slump test for workability assessment, and ii) compressive strength, to determine the strength after its hardening at 1 day, 3days, 7days, and 28 days. For workability cone test was adopted. Cube specimens of standard size were cast and cured in water up to testing age for average compressive strength. Entire the tests were conducted in the Materials laboratory at Civil Engineering Department, QUEST Nawabshah. The results are tabulated and discussed in the following sections.

2. LITERATURE REVIEW

2.1 Rice Husk Ash

The outer coating of the rice grain that is removed in the milling procedure is the rice husk. It is sometimes referred to as rice hull also and is a by product of rice milling. Rice husks account for about one-fifth of the world's 600 million tonnes of rice produced each year. From milled paddy, nearly 78% of the production is received in rice, broken grains, and bran while the remaining 22% of rice's weight is acquired as a husk. Rice husk contains a significant amount of Silicon. Hence it is classified as a pozzolanic substance. Pozzolanic materials are those that behave as siliceous and aluminous compounds. At room temperature, RHA will chemically react with calcium hydroxide during the hydration stage of Portland cement and when exposed to moisture (Boontawee et. al. 2018; Balapour et. al. 2018). This husk is used for producing steam during the parboiling process through burning. It is reported that nearly 22kg husk is produced when 1000kg of rice are processed. Pakistan produces 5.5million tones of rice resulting in 1.1 million tonnes of husk production. The Rice Husk Ash (RHA) is a powder-formed material that is obtained by burning of rice husk. RHA is a potential pozzolanic substance that may be blended with cement to obtain durable concrete.



Figure 1. The Categories of AI

Adding the RHA in cement improves the early strength of the concrete and Calcium Silicate Hydrate (CSH) gel surrounding the cement particles. This gel is dense and less porous, potentially increasing the strength of the concrete against cracking. There have recently been significant attempts globally to use local and waste materials in concrete, such as rice husk. If rice husk is properly burned and suitably pulverized, it produces ash that may be utilised as a cement substitute.

2.2 RHA Properties

The high silica content of rice husk ash (RHA) in non-crystalline or amorphous silica depends on burning temperatures; 95 percent silica may be generated after heating at 700° C for 6 hours. Rice absorbs silica from the soil and integrates it into its structure as it grows. External layer in which the grain of rice is wrapped is known as rice husk and it contains high concentration of silicon i.e. up to 85% which is beneficial to pozzolanic activity. RHA has particle sizes ranging from 5 to 10 micrometres, a specific gravity of 2 to 2.4, and a particular surface area ranging from 20 to 50 m²/kg. When rice husk is burnt into ash, it takes on mineral admixtures' physical and chemical properties. Chemical and physical properties of the RHA are summarized in Table 1.

Chemical Constituent	Range (% by mean)
SiO ₂	82.5 – 97
Fe ₂ O ₃	0.54
Ca O	0.1-1.31
Mg O	0.01-1.96
K ₂ O	0.1-2.54
Na ₂ O	0.01-1.58
P ₂ O ₅	0.01-2.69
Si O ₃	0.01-1.23
Carbon	2.71-6.42
Physical Property	Value of RHA
Fineness (m ² /kg)	360
Loss of ignition	5.9
Bulk density (Kg/m ³)	96-160
Specific Gravity	2.0

Table 1. Chemical and Physical Properties of RHA

To maintain the required quality level of the ash, appropriate incinerator or furnace must be used, and proper grinding method should be adopted. The majority of the ashes exhibit a lesser loss of ignites property. RHA has lower specific gravity than OPC and a significantly higher fineness than OPC, crucial for producing denser and more lasting concrete.

2.3 RHA application in concrete

RHA in recent years has gained attention by the researchers for using as supplementary cementitious material due to the high amount of amorphous silica content available in RHA. Various researchers have conducted the experimental work to assess the adaptability of RHA as a partial replacement of cement and reported that RHA could be show an increase in the strength of concrete up to 30% of cement replacement (Ganesan et. al. 2008; Zareei et. al. 2017). However, the use of traditional RHA directly is unfavourable because the traditional RHA has shown variability in chemical and mineralogical characteristics, which leads to the need of testing characteristic of RHA before using in concrete.

The application of RHA has resulted in several benefits, such as workability. RHA effects on the rheological characteristics, strength, finish, porosity, durability and cost of the concrete (Mehta and Monteiro, 2017). Research work shows that the use of RHA in concrete enhances the workability of fresh concrete and the strength and durability of hardened concrete, in addition to leading to sustainable concrete and improving the durability of hardened concrete (Bapat, 2012; Shetty, 2005). RHA's increased water needs have typically been attributed to their particle size and surface area. On the other hand, Givi et al. (2010a) and Givi et al. (2010b) discovered that the application of RHA decreases the water requirement of mixes. The workability of cement partially replaced by RHA is enhanced, amplified with coarser RHA particles (average particle size of 95m) and raises the replacement ratio (Bui, 2001).

In contrast, some researchers argued that increasing RHA quantity reduces the concrete workability due to absorptive property (Kartini, 2008). Also, the fineness of the particles causes' high demand of water to the wet area of RHA, reducing workability (Ganesan et. al. 2008; Habeeb and Fayyadh, 2009). Besides this, the irregularity of the form of RHA particles also has a significant influence on the workability of the concrete (Srinivasreddy et. al. 2017). In assessing the performance of the concrete, one of the most critical parameters is the strength of the concrete. Strength is the material's capacity to withstand stress without failing (Neville and Brooks, 1987). In general, the concrete's compressive strength depends on w/c ratio, mixture composition. The rate of strength growth at early ages is proportional to the rate of cement hydration. Portland cement contributes significantly to strong growth in the early stages, up to around seven days, after which it decreases significantly. As a result, the addition of RHA to concrete influences all aspects of its characteristics. RHA reacts as a cementitious material in the concrete.

RHA is a strong pozzolanic material and this pozzolanic activity occurs due to silica content. An investigation of the effect of RHA and OPC-based concrete demonstrated that by replacing 30% of the weight of OPC with RHA for 28 days, it is feasible to lower the temperature (Mehta and Pirtz, 1978). Another study investigated temperature range required to burn rice husk for obtaining the appropriate pozzolanic result where it was found that if the rice husk is burned at the optimal temperature, it is possible to substitute up to 40% of the cement with RHA with no noticeable difference in compressive strength compared to the controlled mix (Al-Khalaf and Yousif, 1984). Ryan (1999) performed an experimental investigation on the durability of concrete by studying the effect of sulphate attack

and chloride ion penetration. Skalny et. al. (2002) argued that concrete subjected to sulphate assault experiences a progressive and significant reorganisation of its internal microstructure. These modifications have a direct impact on the material's technical characteristics. Oyetola and Adullahi (2006) calculated compressive strength of sand-crete block prepared by replacing cement with RHA up to 80% and reported increase in compressive strength of the 'OPC/RHA sand-crete' specimens with the age and decrease with the increase in RHA amount and for a specific mix, it is reached an ideal replacement level of 20 per cent. The need for water is also high as the amount of rice husk rises. Prasad et al. (2006) researched cement concrete and found that cement mixed with sodium sulphate settings are superior to magnesium sulphate. The environment of magnesium sulphate is harder than the environment of sodium sulphate. Use of the RHA in substitution of cement showed that Rice Husk Ash's ideal substitution of cement was almost 10% working and resistance (Krishna et al. 2016).

Study of hydration process and the development of compressive strength of concrete using RHA showed that the RHA particle size affected compressive strength for 28 days but replacement content took over for 91 days while the addition of hydration kinetics, the porous structure of RHA particles has enabled its function which enhances compression at a late age (Vieira et. al. 2020). The compressive strength of the concrete using RHA as substitute for magnesium sulphate solution by replacing cement with rice husk ash at the rate of 0%, 5%, 5%, 10, 15%, and 20% by the mass of cement for the mixture. The findings revealed that with the addition of magnesium sulphate, RHA improves the concrete's durability (Sarma et al., 2014). Another study showed that concrete with use of RHA for replacing ordinary Portland cement can increase the compressive strength of the concrete up to the dosage of 10% of RHA of the cement (Ahsan and Hossain, 2018).

Experimental work on studying the properties of mortar and concrete using rice husk ash as partial substitution of cement revealed that RHA consumes more water than concrete without RHA. Also, this replacement of cement up to 10% of RHA is acceptable (Christopher et al., 2017). Another study showed that 25% of the RHA can be used in concrete as a substitute for cement, giving the same results of concrete without RHA and reduced environmental damages (Khan et al. 2012). Kishore et al. (2011) performed experimental work for studying mechanical behaviour of the concrete by replacing cement with RHA with the proportion of 0, 5, 10 and 15 percent while M40 and M50 grade mix of RHA was prepared to determine compressive strength on 7, 28 and 56 days. From this experimental work, the researchers concluded that the workability is reduced to 27% when RHA is added while optimal use of RHA in achieving sufficient strength of the concrete is 10%. Beyond this, the concrete gradually loses its compressive strength. The use of RHA in concrete also has a positive effect showing the resistance to Sulphate attacks. This was demonstrated through experimental work carried out by adding RHA with cement as a partial replacement, where it was found that concrete using RHA up to 20% exerts substantial resistance to sulphate attack (Jaya et al. 2018). However, many researchers have argued that use of RHA in concrete consumes more water than OPC based concrete. Hence a study focused on assessing w/c ratio in concrete using RHA together with OPC by testing 150 samples and it was reported that optimal water/binder ratio is 0.50 to 0.55 and an optimal amount of

RHA for achieving a workable concrete is 15% (Abdullahi et al., 2017).

Madandoust et al. (2011) investigated the mechanical characteristics and durability of rice husk ash concrete. The results showed that replacing 20% of the cement with RHA enhanced durability and homogeneity but did not raise the early age compressive strength of concrete. Concrete containing RHA, on the other hand, has higher compressive strength at later ages. RHA was also tried as a cementitious material in recycled aggregate concrete (Rattanachu et al., 2020). The results showed an increase in compressive strength when 20% of OPC was replaced with finely ground rice husk ash. However, the increase in the strength in concrete was observed at later age's i.e., after 60 days. This is due to a secondary reaction between the supplementary cementitious material and water in the presence of the alkali produced during the hydration of cement as the primary reaction.

Tashmia et al. (2005) studied the mechanical characteristics of RHA concrete. The cement replacement with 5% and 10% of RHA was adopted. Samples were prepared and tested for average compressive strength, flexural strength, water absorption, and modulus of elasticity. The authors reported that due to the addition of RHA, water absorption reduced, particularly at later ages. Also, increase in compressive strength was noted, but a little bit of effect in the case of splitting flexural strength due to RHA was recorded. However, the modulus of elasticity remained unchanged. Several researchers have reported different dosages suitable for the replacement of cement with RHA achieving sufficient strength and improved workability. Rasoul et al. (2017) cited that Isaia et al. (2003) discovered 50% RHA substituting cement having the identical strength value to that of ordinary concrete. Still, Ganesan et al. (2008) proposed the maximum replacement level of OPC with RHA to be 30% beyond which deterioration in the concrete performance was expected. Another researcher Ettu et al. (2013) limited the RHA dose only at 15% based on his study and the type of RHA used and concrete produced then. Whereas, Safiuddin (2010), optimized the RHA dosage ranging between 10% to 20%. It is worth noting that a bare minimum value of RHA being 5% only is also recommended by Leong (2015) for the improved strength of concrete. Even a few researchers have reported adverse effect of RHA on concrete's strength irrespective of the dosage. At any dosage, the decreased strength compared to normal concrete is reported by Madandoust et al. (2011) and Marthong (2012). This variation in the optimal value of RHA dosage has been attributed to many factors but the dominating factors are outlined as; fineness of the silica particles resulting in their huge surface area and the particle properties such as shapes (spherical to irregular ratio) (Safiuddin et al., 2010).

Based on above literature review, it can be concluded that the Rice Husk Ash (RHA) is obtained by the burning of 'Rice Husk', an agricultural waste of Rice crop during rice processing. RHA possesses siliceous properties which tends it to be a cementitious material. Likewise, other cementitious materials, its effectiveness depends on many factors such as the source of the Husk, the burning temperature, fineness of the particles, replacement level of cement with RHA, type of concrete, water-binder ration, curing regime and age etc. It is, therefore, the literature reports a lot of discrepancies and staggering results based on the studies carried by different

researchers. Pakistan is an agricultural country producing rice crop as one of the main crops. Thus, a substantial amount of rice husk is also produced during the rice processing. This husk either is used as fuel in the rice mills, leaving rice husk ash again causing environmental issues or husk directly is dumped in open areas, leading to the scarcity of the dumping space and environmental problems. This led to the idea to undertake the experimental study to understand the effect of rice husk ash by burning the husk in the laboratory and husk obtained from the local areas. For this study, rice husks were collected from Nawabshah city.

3. MATERIALS AND METHOD

The performance of concrete in a hardened state depends on the ingredients and their properties, mix proportion, manufacturing process and method, curing regime and age etc. Therefore, the selection of materials, concrete preparation and production practices and procedures and other parameters should be very carefully carried before starting any experimental investigations. We used Mix proportions by weight and the RHA (Rice Husk Ash) at dosage Levels of 0%, 5%, 10%, 15%, 20%, 25%, and 30% by weight of binders to replace OPC. The mix proportion was calculated and is presented in Table 2.

S #	Batch	Dosage of RHA
1	B – 1	0%
2	B – 2	05%
3	B – 3	10%
4	B – 4	15%
5	B – 5	20%
6	B – 6	25%
7	B – 7	30%

Table 2. Details of the batches of specimens cast and tested

In this study, cubes of standard size 100mm x 100mm x 100mm were cast for testing purposes. All cubes were cured in water for 1, 3.7, and 28 days. The study was carried to study the workability and compressive (cube crushing) strength at different ages. Workability refers to the ease with which concrete may be poured and compacted without segregation. ACI defines workability as "the quality of newly mixed concrete or mortar that dictates the ease with which it may be mixed, put, consolidated, and finished in a homogeneous condition" (Zulu, 2017). Workability is an important characteristic of concrete that is connected to compaction and strength. The ideal workability varies depending on the kind of concrete.

4. RESULTS AND DISCUSSIONS

The cubes were tested for examining the two properties as workability and compressive strength. For assessing the workability, slump test was conducted in each case before the casting of the specimens. Table 3 shows the average values of the slump cone test, which are graphically represented in Figure 1.

The values in table 3 show that the workability decreases due to the addition of the RHA as a partial cement replacement. The maximum slump value obtained is 64 mm in the case of

ordinary concrete, and it is decreased to 7 mm when 30% RHA is added, which is just 11% of the slump value of concrete without RHA.

% of RHA	Average Slump values (mm)
0%	64
5%	54
10%	41
15%	14
20%	11
25%	9
30%	7

Table 3. Workability at different percentages of RHA

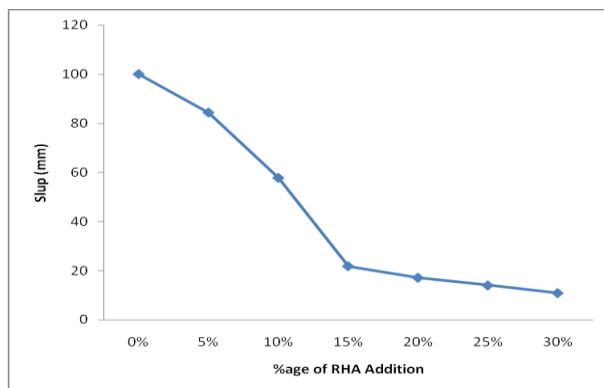


Figure 1. Workability of RHA Based Concrete

It is also obvious from the slump values determined that the reduction in slump value becomes more prominent with the further increase in RHA dosage and it is not proportional to the dosage of RHA. For example slump value of the concrete at 5% RHA is dropped to 54mm, followed by 41mm, 14mm, 11mm, 9mm, and 7mm at 10%, 15%, 20%, 25%, and 30% of RHA respectively and these are 58%, 22%, 17%, 14%, 11% of the concrete without RHA. It is also notable that at the higher dosage of RHA i.e. 15% and beyond, the workability becomes the bare minimum to be considered as workable concrete. It infers that the use of RHA causes un-proportional alteration in the water-binder ratio of concrete. Thus, the dosage of RHA is decided with particular care while considering the desirable workability of the concrete using RHA.

The tests for determination of compressive strength were conducted in the FORNEY machine with gradual application of the load. Before the testing, the surfaces of the specimens were cleaned with dry cotton to avoid water and any substance at the surface. In each case, five specimens were tested to have the average values. The specimens were loaded up to their crushing. Careful and close observations were made during the testing, and the readings were noted very carefully immediately after the crushing of the specimens. At the specimens, it was also observed that whether the failure was catastrophic and violent or otherwise. Almost in all the cases, it was identical, being sudden and violent and crushed completely at the time of failure. The recorded readings were tabulated, and calculations were made accordingly, followed by their analysis. The results and their analysis, and comparisons are presented in the tables.

Compressive strength of the concrete with different dosages from 0% to 30% replacement is shown in Table 4.

Batch	RHA	Average Compressive Strength (MPa)	Difference %
Curing Age: 1 Day			
1	0%	22.13	
2	5%	23.07	4.22
3	10%	23.00	3.92
4	15%	20.27	-8.43
5	20%	17.27	-21.99
6	25%	15.60	-29.52
7	30%	15.27	-31.02
Curing Age: 3 Days			
1	0%	24.43	
2	5%	24.40	-0.14
3	10%	21.13	-13.51
4	15%	19.23	-21.28
5	20%	15.80	-35.33
6	25%	14.73	-39.70
7	30%	12.20	-50.07
Curing Age: 7 Days			
1	0%	20.67	
2	5%	20.60	-0.32
3	10%	20.53	-0.65
4	15%	17.97	-13.06
5	20%	16.43	-20.48
6	25%	14.53	-29.68
7	30%	13.87	-32.90
Curing Age: 28 Days			
1	0%	24.63	
2	5%	24.30	-1.4
3	10%	24.47	-0.7
4	15%	21.07	-14.5
5	20%	15.33	-37.8
6	25%	14.10	-42.8
7	30%	13.33	-45.9

Table 4. Compressive Strength of RHA Based Concrete

The results of the compressive strength in table 4 depict that;

- The compressive strength decreases with the addition of RHA.
- The increase in the dosage of RHA further decreases the compressive strength
- The maximum compressive strength achieved at 28 days is 30.8 MPa.
- The minimum compressive strength achieved at 1 day is 61.5% of its 28 days strength followed by 3days 7 days, respectively
- It is notable that as the dosage of RHA increase, the strength development at the early stage is less than the ordinary concrete
- For example, at 1 day, the compressive strength achieved is 18.9 MPa which decreases to 15.5 MPa if 10% of RHA is used and 7 MPa when RHA addition reaches to 30%.
- The decrease in compressive strength due to RHA is abrupt beyond 10% of RHA. Therefore, it infers that in terms of average compressive strength, RHA may be optimum as 10% replacement and curing be continued up to 28 days in order to achieve the adequate hydration of RHA to attain the appropriate strength.

As the results show remarkable effect of RHA on average compressive strength of concrete in general but more in case of the concrete with higher dosage of RHA. It is worth noting that in most of the cases, RHA has exhibited adverse effect when compared to the ordinary concrete without RHA. This was expected because during this study the Rice husk was obtained from the local Rice mill and was burn in the laboratory, the temperature might have been moderate at about 2000C. However, the temperature should quite higher about 5000C to have completed burning that siliceous behaviour may have been produced. Nevertheless, the current study also shows that even RHA obtained at low temperature burning may be used up to 10% as cement replacing material with a very marginal reduction in strength.

5. CONCLUSION

Experimental investigations carried on the concrete with RHA in terms of workability and average compressive strength are concluded as under.

- The addition of RHA as supplementary binding material changes both the properties studied; workability and compressive strength.
- The workability of concrete was measured by adopting slump cone test and was measured in millimetres (mm). Workability of concrete due to RHA decreases in all the cases of the RHA replacement level. The decrease was not proportional to the increase in the dosage.
- A smaller dosage the extent of decrease was low as compared to the drastic decrease in the workability beyond the 15% of RHA in concrete. it almost reached to non workable concrete at 25% and 30% RHA dosage.
- The maximum workability obtained was 64mm in ordinary concrete without RHA and it reached only 7mm for concrete with 30% RHA. This shows that the addition of RHA in concrete should be carried carefully in terms of the workability and necessary admixtures like plasticizers may be used to achieve minimum required workability of concrete.
- The average compressive strength of concrete of ordinary concrete at 28 days was found to be maximum being 24.63 MPa and that of the same concrete and at same age with 5% and 10% RHA was observed to be 24.43 MPa and 24.47 MPa respectively. The difference is negligible being 1.4% and 0.7% only. This infers that RHA may be used as cement replacement in concrete up to 10% by weight of cement.
- Beyond 10% of RHA substantial decrease is observed in average compressive strength being 14.5%, 37.8%, 42.8% and 45.9% in case of RHA replacement being 15%, 20%, 25% and 30% respectively. However, it is worth noting that, the concrete with RHA 30% is also applicable in relatively low strength applications. Also during this study workability was maintained because of not using any plasticizer. It is expected that keeping all other aspects same, if plasticizer be used to maintain the workability, the

compressive strength may be reaching to that of ordinary concrete. This aspect needs to be investigated in future.

- On the basis of the experimental study carried out and the results cited above, it may be concluded that, the 10% RHA may be considered as optimum level without compromising both the aspects studied i.e workability and average compressive strength of concrete which were almost equal to those of the ordinary concrete without RHA.

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